# Portfolio Risk Calculation

SPY-TLT Portfolio and AAPL-MSFT Portfolio

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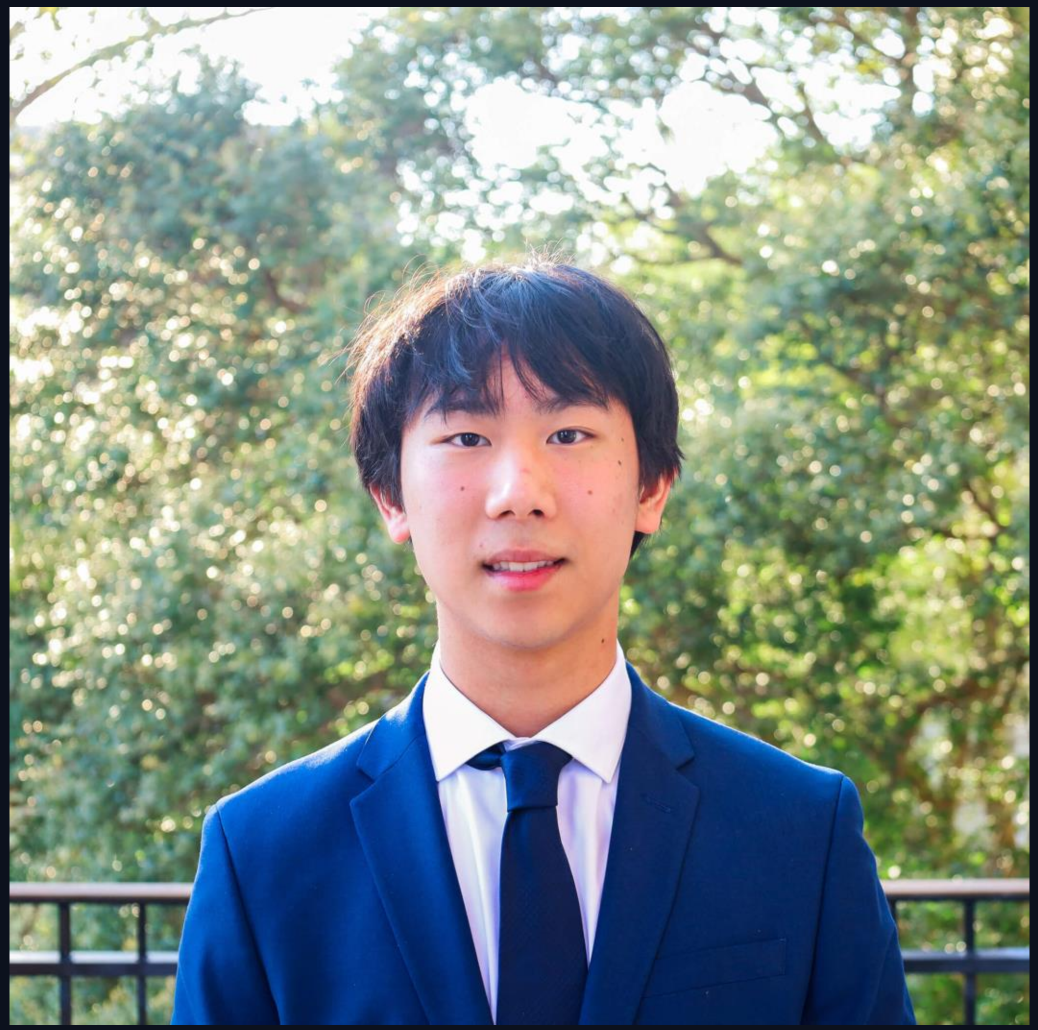
# Meet The Team

Zabe Nash

Sina Tegene



Max Fan

Gabe Pringle

SESSION

# Agenda

- Project Overview
- Asset Class Overview
- Visuals
- Conclusion

# Project Overview

## Two Portfolios

- 60% SPY / 40% TLT (Traditional Diversified)
- 60% MSFT / 40% AAPL (High-Growth, High-Beta)

## Methodology

- Created visuals to highlight differences in portfolio behavior
- Computed portfolio and individual standard deviation and beta
- Compare the portfolio's risk to the individual components

# S&P 500 Index (SPX)

**S&P 500 Index (SPX) – April 29, 2025**

- **Closing Price:** 5,560.83
- **Daily Change:** +32.04 (+0.58%)
- **52-Week Range:** 4,835.04 – 6,147.43
- **Year-to-Date Performance:** Down 5.5%

The S&P 500 marked its **sixth consecutive daily gain**, driven by stronger-than-expected corporate earnings and easing trade tensions. Notable contributors included Honeywell International and Sherwin-Williams, both of which reported robust profits and raised future expectations.

# iShares 20+ Year Treasury Bond ETF (TLT)

- **Closing Price:** $90.20
- **Daily Change:** +$0.76 (+0.85%)
- **52-Week Range:** $84.89 – $101.64
- **30-Day SEC Yield:** 4.66%
- **Effective Duration:** 15.88 years
- **Year-to-Date Total Return:** +3.29%

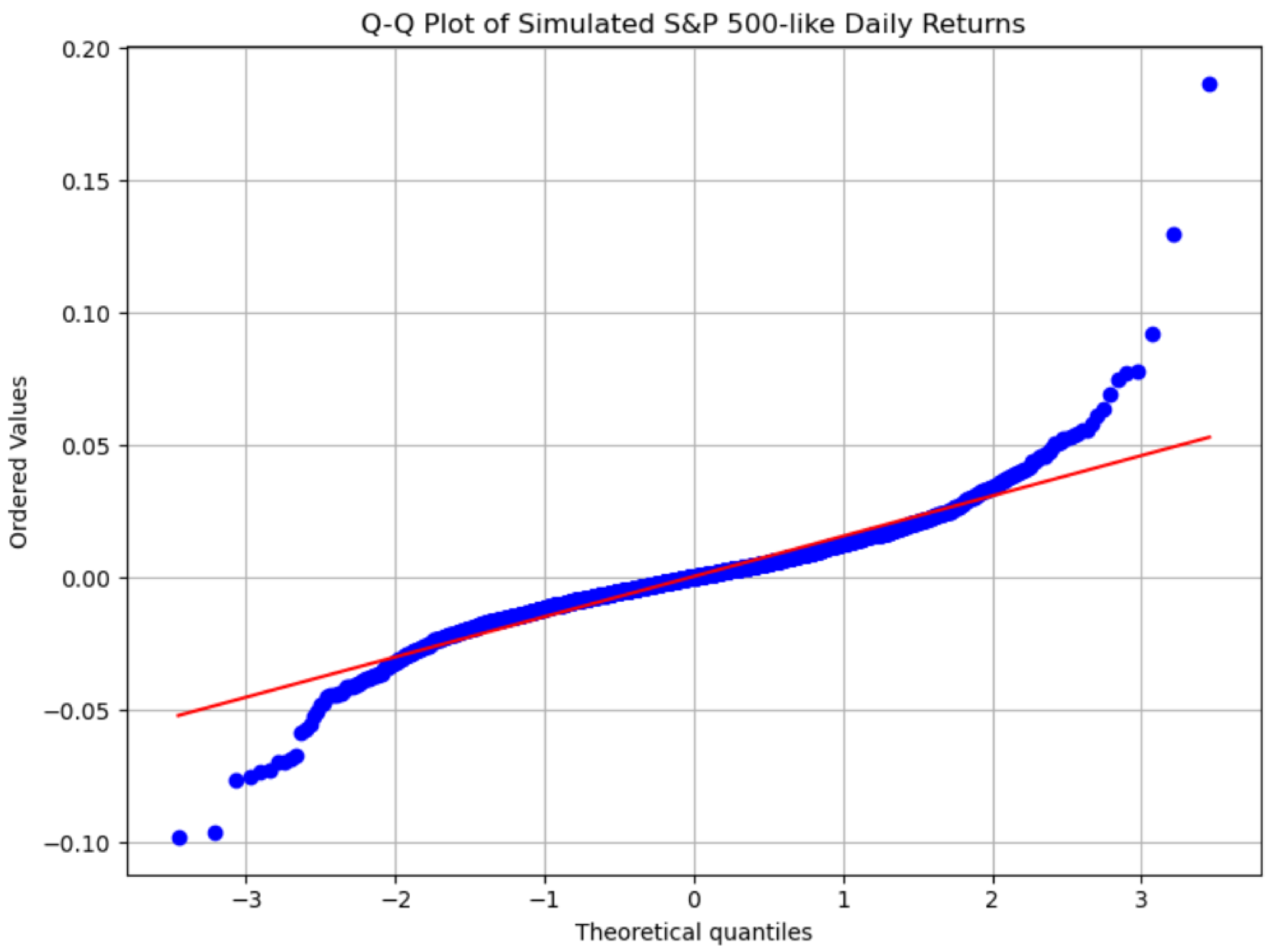
**Experienced increased investor interest recently due to several factors**

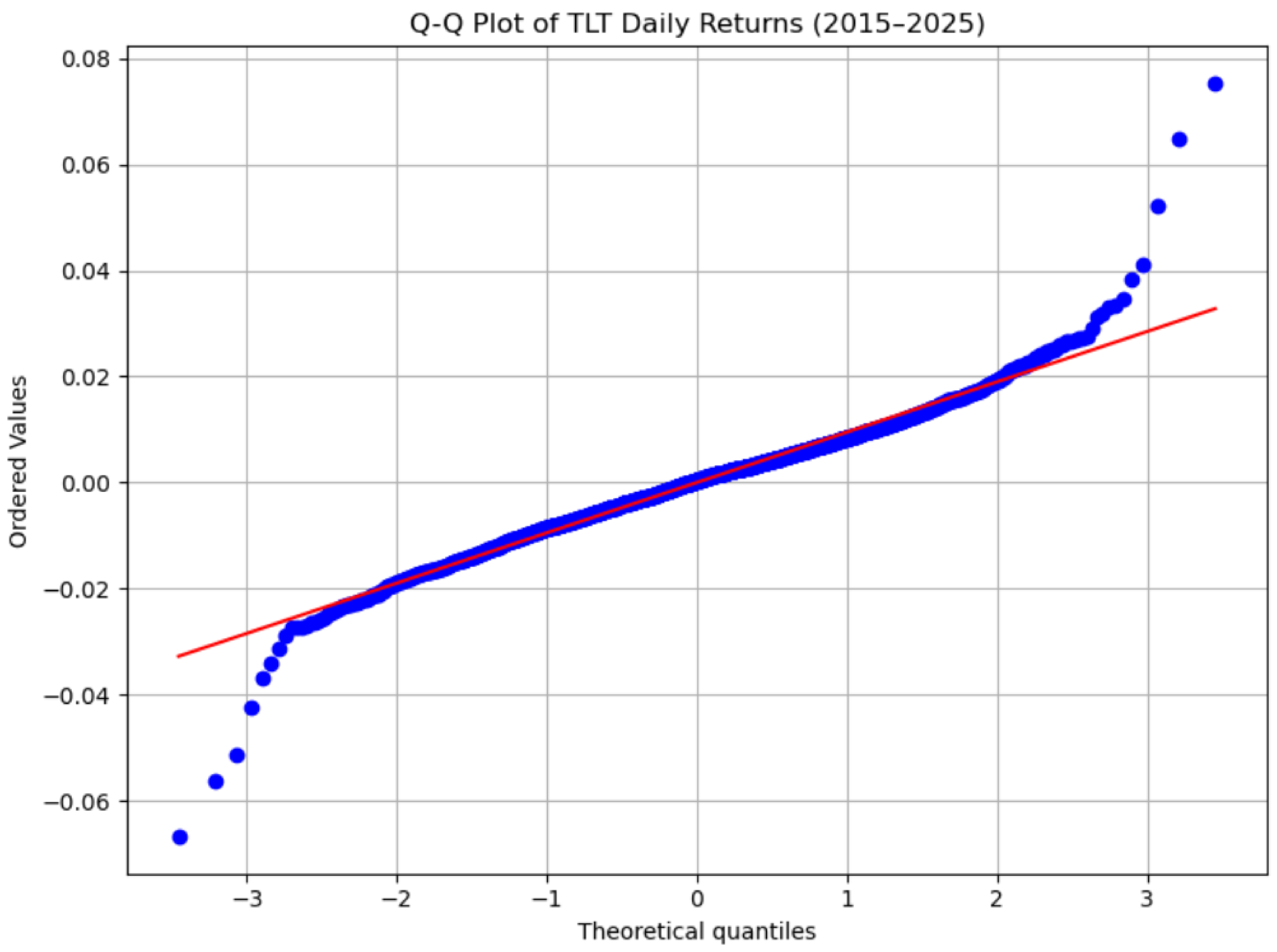
**Inflation Concerns:** Investors are closely watching the upcoming Personal Consumption Expenditures (PCE) inflation report. A lower-than-expected inflation reading could bolster bond prices, benefiting TLT.

**Treasury Department's Borrowing Plans:** The U.S. Treasury is set to release its quarterly refunding announcement, outlining borrowing plans for the upcoming months. While the Treasury is expected to maintain the current size of note and bond auctions, any changes in forward guidance could impact long-term yields and, consequently, TLT's performance.

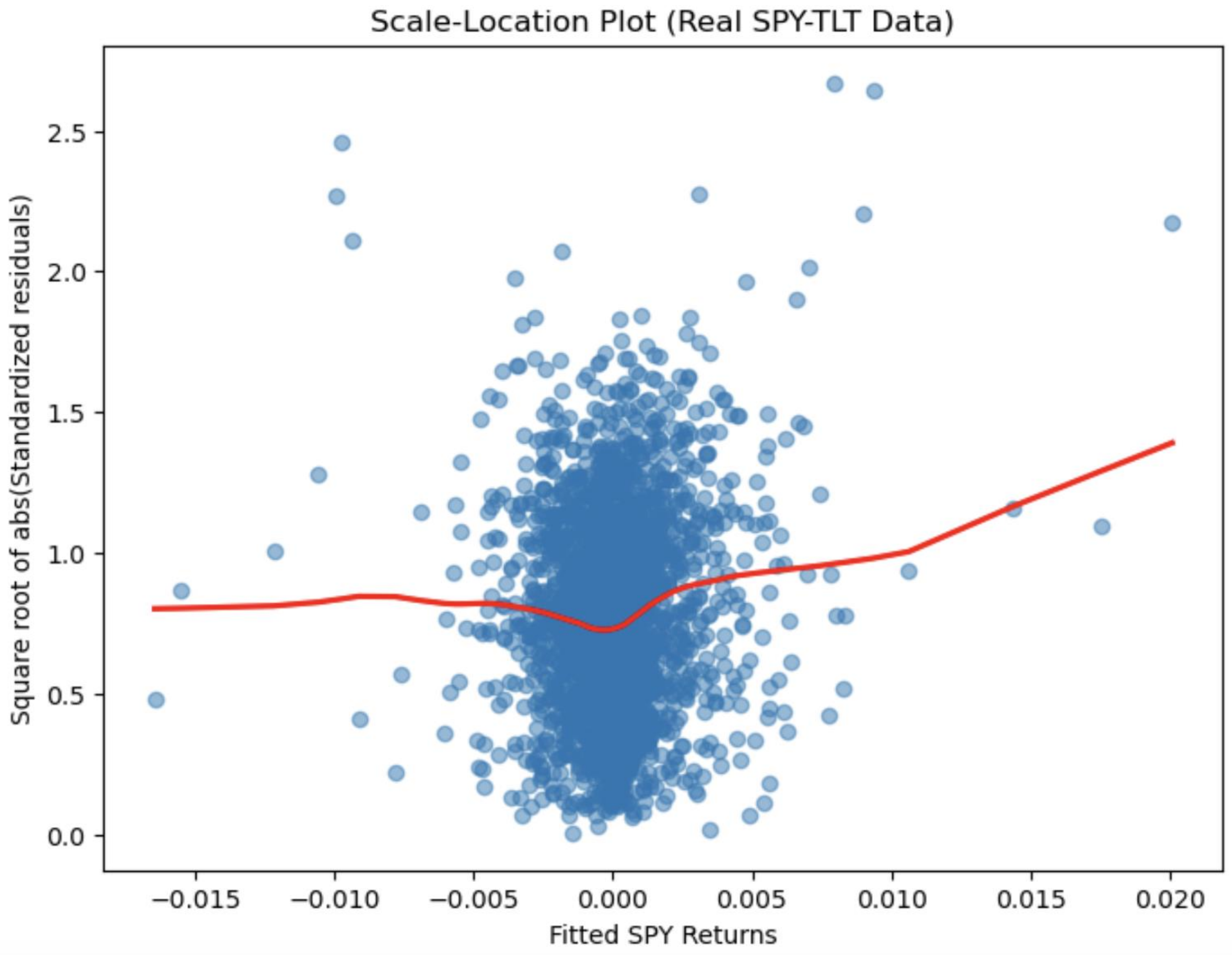
**Market Volatility:** Recent fluctuations in Treasury yields, driven by trade policy uncertainties and economic data, have led to increased volatility in long-term bond ETFs like TLT.

# Q-Q Plot of SPY and TLT Daily Returns



# Scale Location Plot SPY-TLT




- Shows predicted returns in relation to size of the prediction errors
- Blue Dot is a daily return from 2015-2025
- Redline shows trend in size of errors
- Model does well in a flat market
- Model does worse in a volatile market

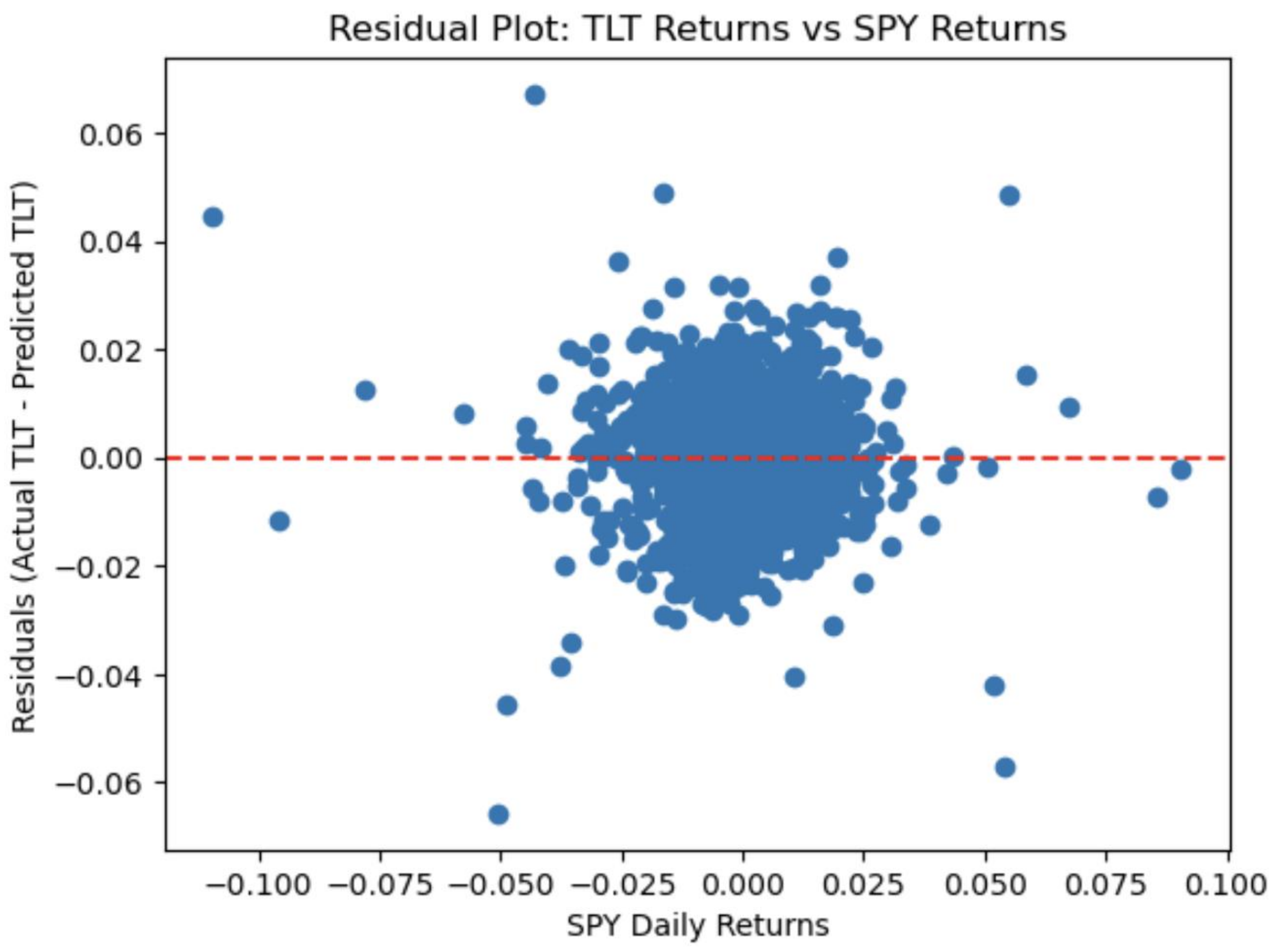
# Residual Plot: TLT Returns and SPY Returns

### What the shape shows

- The plot is centered around **0**, meaning the model has no consistent bias (which is good).
- There's **no clear pattern** in the residuals - they're randomly scattered.
- This randomness suggests the **linear model is a good fit** - no major non-linear patterns are left in the data.

### Interpretation

This plot supports the idea that while SPY and TLT are related, the relationship isn't perfectly predictive and that's why diversifying across them can reduce portfolio risk. The randomness in residuals highlights that they don't always move in lockstep.




**X-axis:** SPY daily returns
 - These are the daily percentage returns of SPY (e.g., -0.05 means 5%)
**Y-axis:** Residuals (Actual TLT return − Predicted TLT return)
 - If a residual is **0**, the model predicted the TLT return **perfectly** for that day.
- If it's **positive**, the model **underpredicted** TLT.
- If it's **negative**, the model **overpredicted** TLT.
**Each dot** = one trading day

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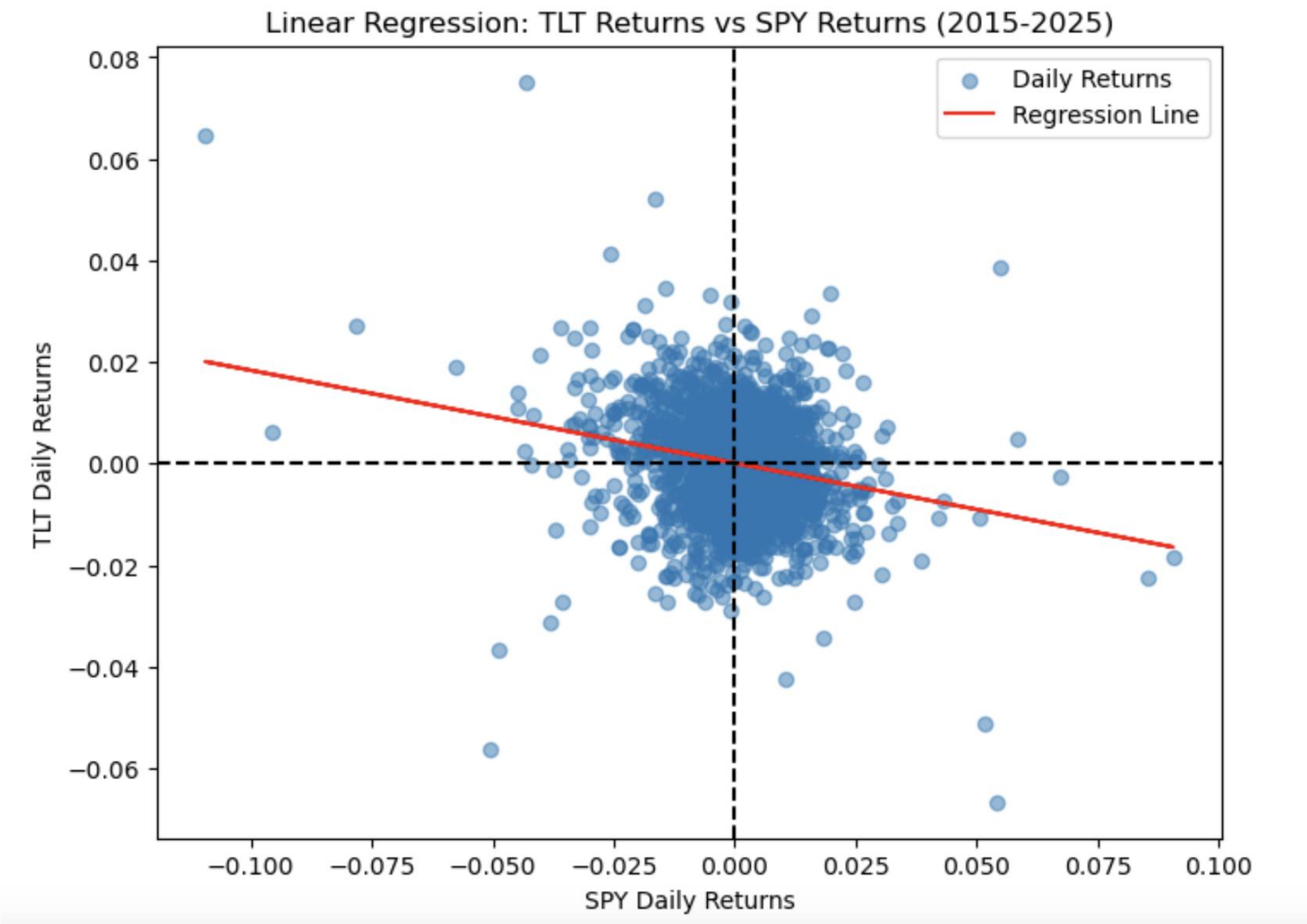
**Purpose of the Plot:** To evaluate how well SPY returns explain the movement in TLT returns using a linear model.

# Linear Regression: TLT Returns vs. SPY Returns




- Correlation Coefficient (r): -.210
- Coefficient of Determination (R^2): 0.044
- SPY Daily Standard Deviation: 0.011
- TLT Daily Standard Deviation: 0.009
- Portfolio Standard Deviation: 0.007
- Beta of SPY (β) = 1.000
- Beta of TLT (β) = -0.183
- Portfolio Beta (β) = 0.527





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# Microsoft Corporation (MSFT)

**(MSFT) – April 29, 2025**

- **Closing Price:** $394.04
- **Daily Change:** +$2.85 (+0.73%)
- **52-Week Range:** $344.79 – $468.35
- **Market Cap:** $3.13 trillion
- **P/E Ratio:** 33.79
- **EPS:** $12.41
- **Volume:** 14.97 million

Microsoft is scheduled to report its Q1 2025 earnings after the market closes on April 30. Analysts anticipate revenue of $68.44 billion, marking a 10% year-over-year increase, and net income of $23.94 billion or $3.21 per share. The Intelligent Cloud segment, particularly Azure, is expected to show strong performance with 18% growth to $26.13 billion. Despite a nearly 7% decline in its stock value year-to-date, analysts maintain a bullish long-term outlook due to Microsoft's AI capabilities.

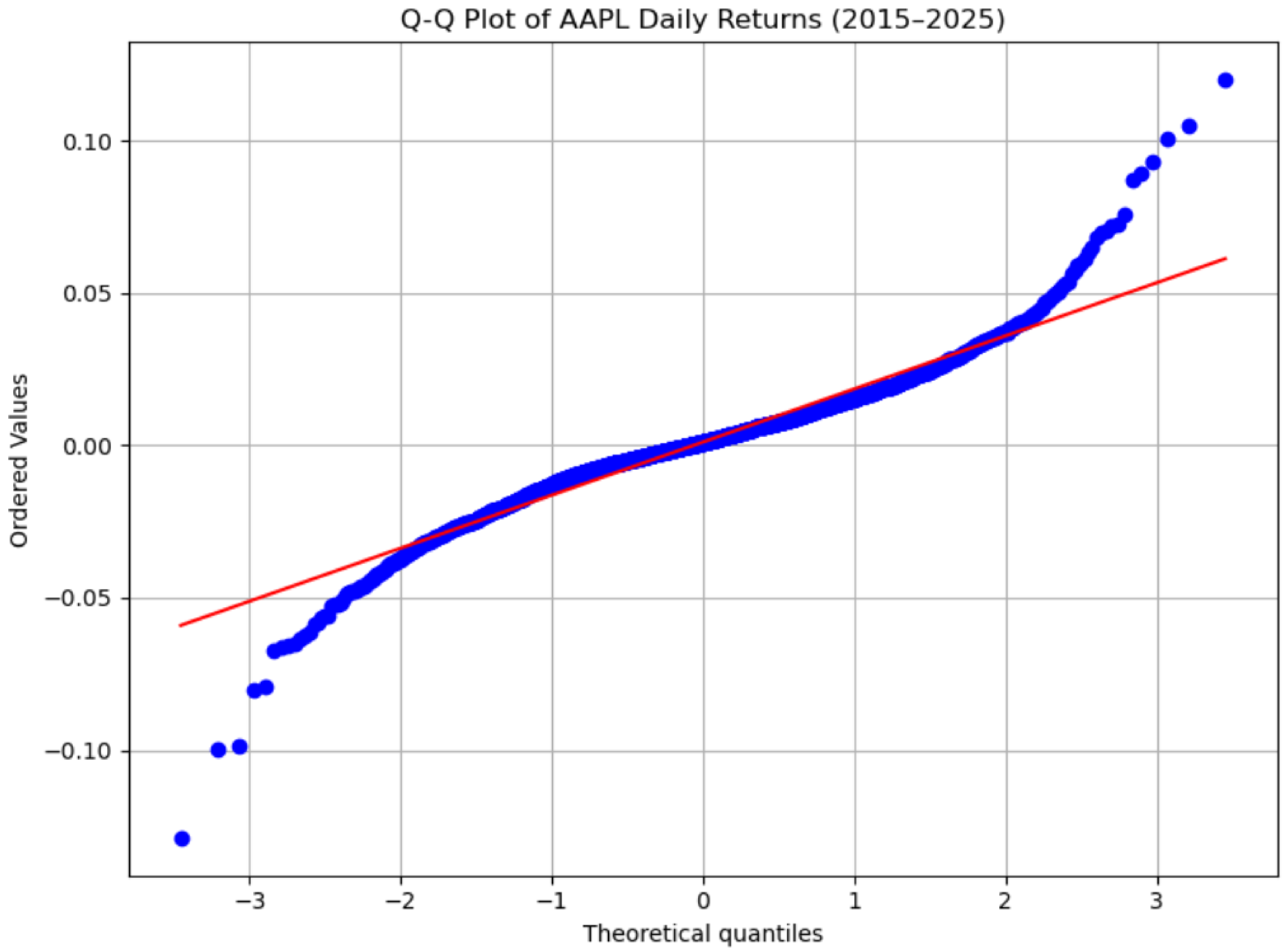
# Apple Inc. (AAPL)

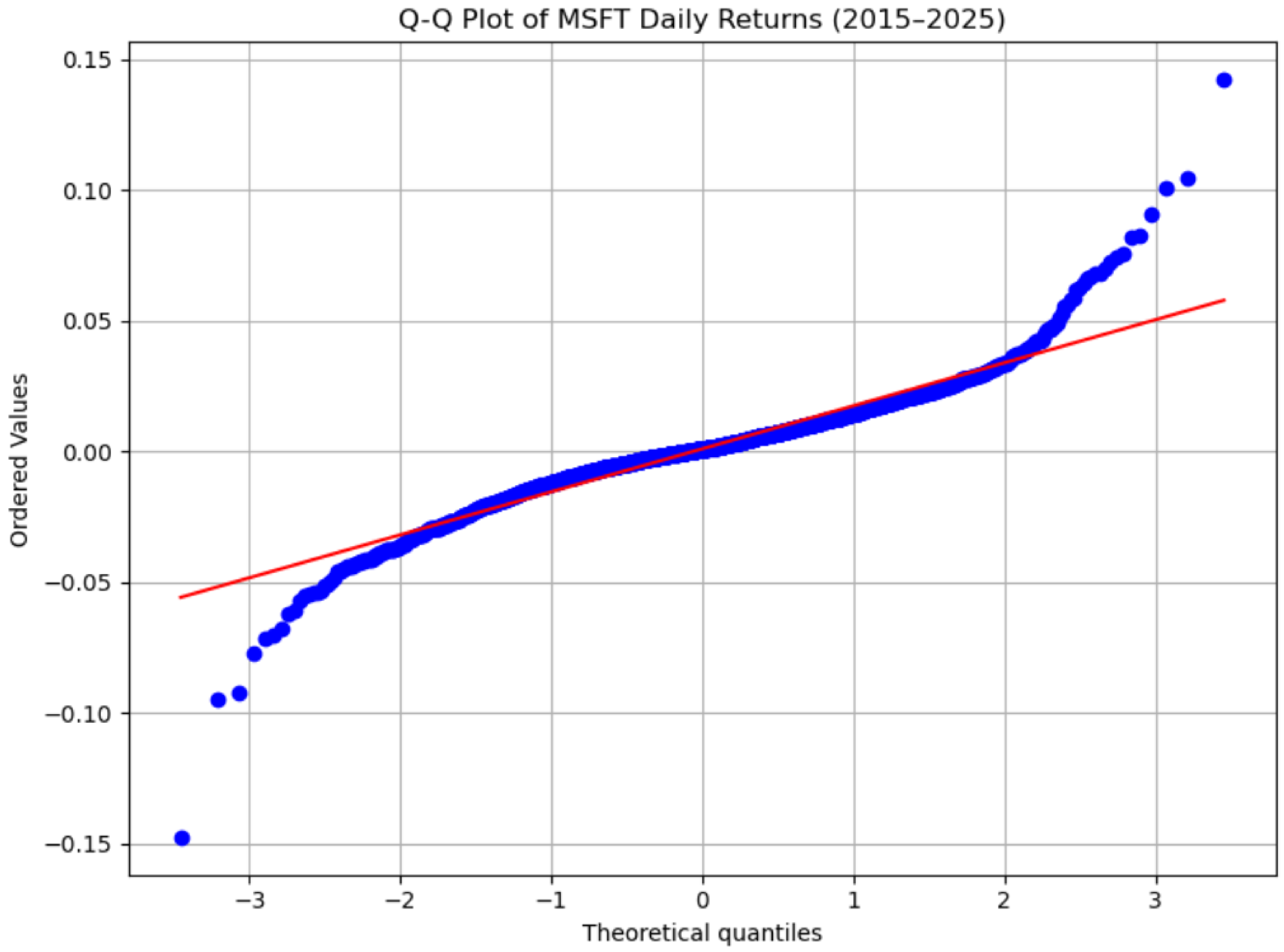
**(AAPL) – April 29, 2025**

- **Closing Price:** $211.21
- **Daily Change:** +$1.07 (+0.51%)
- **52-Week Range:** $169.11 – $260.10
- **Market Cap:** $3.17 trillion
- **P/E Ratio:** 40.27
- **EPS:** $6.30
- **Volume:** 36.83 million

Apple is set to report its Q1 2025 earnings on May 1. Investors are keenly watching for updates on how potential tariffs might affect iPhone sales, as well as developments in the company's AI initiatives. Analysts have adjusted their price targets in anticipation of these factors.

# Q-Q Plot of Apple and Microsoft Daily Returns



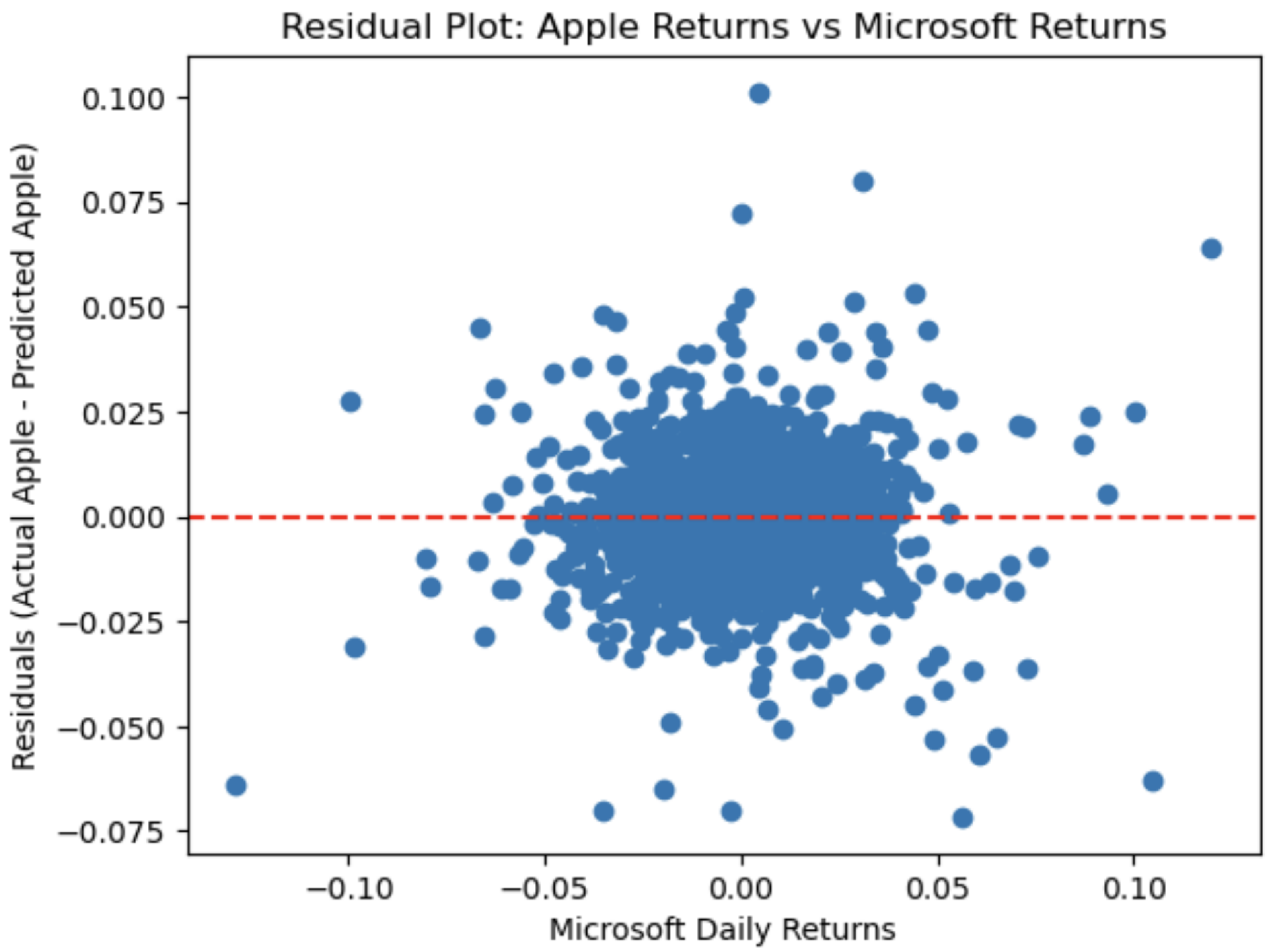
# Residual Plot: MSFT Returns and AAPL Returns

### What the shows

- The residuals mostly fall around the red dashed line (y = 0), indicating no major bias in the predictions.
- There's no obvious pattern in the residuals, which means the linear model captures the main relationship well, and the leftover "noise" is random.
- Most residuals are small and close to zero, suggesting that Apple and Microsoft returns are relatively **strongly correlated**

### Interpretation

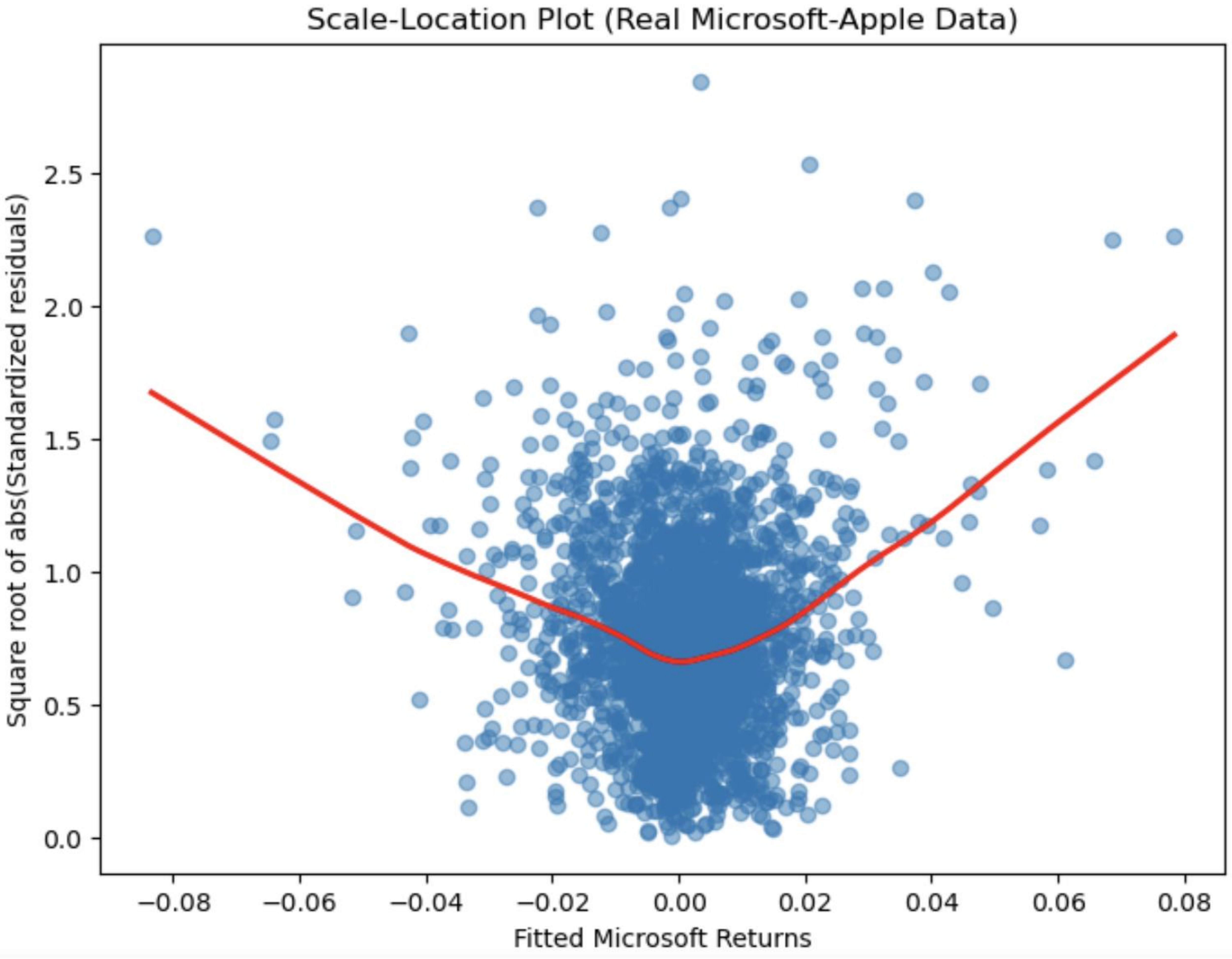
- Unlike TLT and SPY, which are only loosely correlated (diversification benefit), Apple and Microsoft move more similarly, so adding both might not lower portfolio risk much.
- This is a good way to visually show **correlated tech stocks behave similarly**, and how that affects **portfolio concentration risk**.




**X-axis:** MSFT daily returns
 - These are the daily percentage returns of MSFT (e.g., -0.05 means 5%)
**Y-axis:** Residuals (Actual AAPL return – Predicted AAPL return)
 - If a residual is **0**, the model predicted the AAPL return **perfectly** for that day.
- If it's **positive**, the model **underpredicted** AAPL.
- If it's **negative**, the model **overpredicted** AAPL.
**Each dot** = one trading day

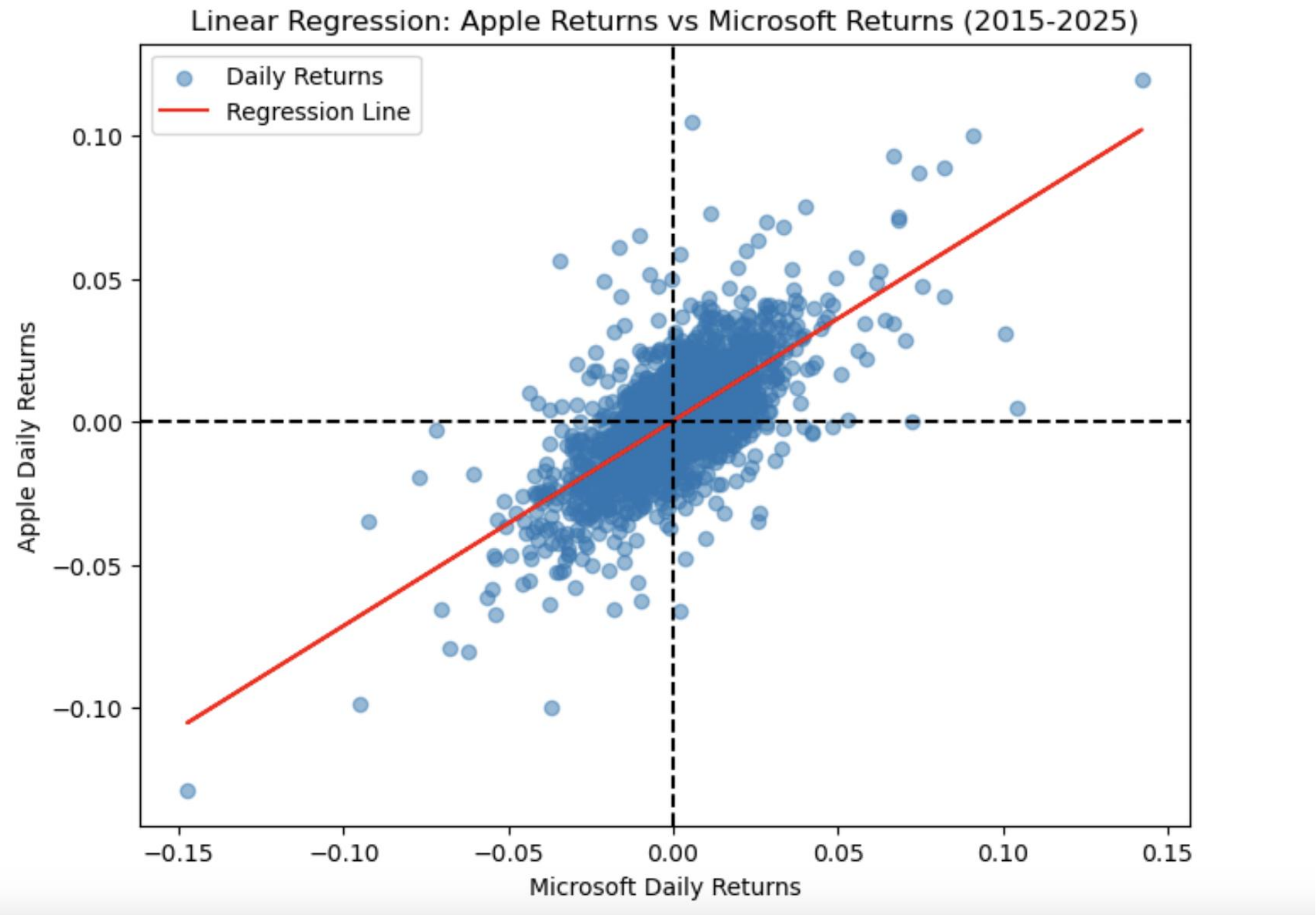
**Purpose of the Plot:** To evaluate how well MSFT returns explain the movement in AAPL returns using a linear model.

# Scale Location Plot (Real Microsoft Apple Data)




- Red U-Shaped Line shows dependence
- Strong positive correlation
- Affected by same news/trends
- Not good for portfolio diversification

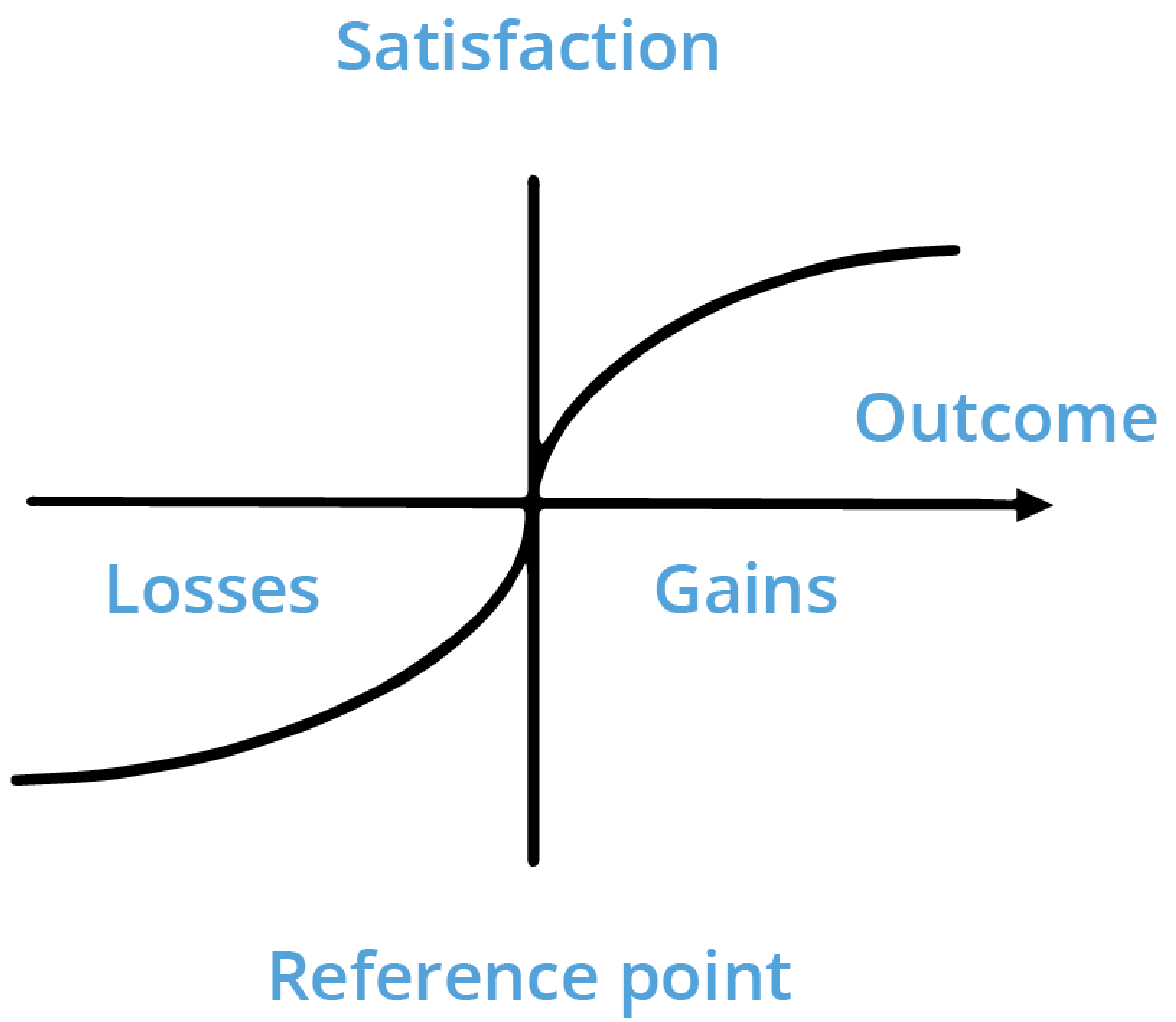
# Linear Regression: AAPL Returns vs. MSFT Returns




- Correlation Coefficient (r): 0.683
- Coefficient of Determination (R^2): 0.466
- AAPL Daily Standard Deviation: 0.018
- MSFT Daily Standard Deviation: 0.017
- Portfolio Standard Deviation: 0.016
- Beta of AAPL (β) = 1.207
- Beta of MSFT (β) = 1.225
- Portfolio Beta (β) = 1.218

# Conclusion

Although a concentrated portfolio of high-beta growth stocks like Apple and Microsoft may offer compelling returns, it also introduces significant volatility and greater sensitivity to market downturns. In contrast, a diversified portfolio that pairs SPY, a broad market ETF, with TLT, a long-term Treasury bond ETF with a negative beta, provides natural downside protection. TLT's inverse relationship with equities allows it to rise or remain stable when stocks decline, reducing overall portfolio volatility and helping preserve capital during market stress. This risk-buffering effect enables more consistent returns over time, even if individual components underperform on a standalone basis. The strength of diversification lies not in maximizing any single asset's return, but in constructing a portfolio where uncorrelated exposures work together to mitigate losses and support long-term resilience - making it the more robust investment strategy.



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